

# Pattern-design Software of Automated Control Systems

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**Abstract**—The method of pattern design of industrial control system (ICS) software has been developed. As objects of lower hierarchy level, the elements of automation were selected. The unified algorithms of data processing and computing control actions were applied for these elements. The data types, data storage and functionality were defined for these objects. In the software of human-machine interface and controller, the correlated structure types were applied. The data blocks in the form of one-dimensional array user-defined type were used for storage. The correspondence of the structure tag to element of array in data block was defined. The data exchange over control system network is independent on the number of sensors and actuators. The template projects for development of software with Siemens hardware and software were created on the basis of proposed method. The structure of the template projects makes it easy to adapt them to the requirements of a particular control system. The template projects greatly reduce time needed for ICS software development and require less programming skills. A universal approach of the proposed method allows using the pattern-design on the automation of various industrial processes.

**Keywords**—automation, control system; software; pattern-design; supervisory control

## I. INTRODUCTION

In many cases a large number of sensors and actuators are used for automation of industrial processes. The settings of these objects are adjusted by means of the commissioning of automation systems and their input/output data is used for industrial process control and monitoring. The complex relationship between these settings and data complicates the development of software for industrial control system (ICS), which requires high qualification of programmers [1, 2].

The use of modern software development tools for controllers and SCADA systems greatly simplifies the process of creating a control system [3-5]. But having a large numbers of sensors and actuators requires much time consuming for development of industrial control system software. It should be noted that the processing and sharing of data through the network between controller user program and human machine interface software occurs in real time [6].

## II. STATEMENT OF THE PROBLEM

In many cases, different control systems differ from each other only in the number of sensors and actuators. In such

cases, it becomes expedient to use template projects that can be easily adapted to the requirements of a particular control system [7]. In this work, the creation of such template projects is proposed to be carried out by means of the pattern design method [8 – 10]. The method is based on the principle of dividing a complex problem into smaller ones. It allows you to rank and analyze the objects in such a way that it is possible to represent the complex interaction of the system elements. The structure and functionality of ICS software should be predetermined in the template project.

## III. RESULTS AND DISCUSSION

Template projects are based on hardware and software from Siemens [11 – 13]. Software of programmable logic controller (PLC) is designed in Step 7 (PLC template project), and performs automatic control of industrial process. Human machine interface software, developed in WinCC SCADA system (HMI template project), is intended for the monitoring and supervisory control of industrial process. At the same time the data exchange over ICS network takes place through the use of built-in WinCC drivers, which greatly increases the reliability of the control system.

In accordance with the patterning method, the same type objects of project, which have the same data processing algorithms and calculation control actions, are applied as the elements of lower hierarchical level. For them, it is necessary to determine the data types, data storage, functionality and organization of exchange between the PLC and HMI stations.

### A. Data Types

In the PLC template project for each type of sensor or actuator, user-defined data type (UDT<sub>k</sub>) is created (Fig. 1). Elements of these structure types are designed to store input/output data and settings of objects.

In the HMI template project for each type of sensor and actuator, the structure types ST<sub>k</sub> are created. The number of elements in structure type of the WinCC project equals to the number of elements in relevant user-defined type of the Step 7 project. Types of these elements match the type of data storage in the memory according to the IEEE standard. To do this, the data types REAL, INT, BOOL are selected in Step 7, and in WinCC – the corresponding data types FLOAT, WORD, BIT.

### B. Data Storage

Data storage of the same type of sensors and actuators in the PLC template project is implemented as a data block  $DB_k$ . It is a one-dimensional array with user-defined type elements. In the HMI template project, the  $M_k$  of same type structure tags  $T_{km}$ , which are combined in appropriate groups of  $TG_k$  tags are created to data storage for each type of sensor and actuator.

For each element of the array in controller user program it should be the structure tag  $T_{km}$  in the human-machine interface application. To do this, as the address of structure tag  $T_{km}$ , the offset of array element in data block  $DB_k$  is selected.

### C. Network data exchange

The application of data block for storage greatly simplifies the organization of change between the PLC and HMI stations, as Step 7 and WinCC have the ability to exchange data blocks [12]. The condition for the exchange of data blocks is a corresponding elements in the structure types in PLC and HMI template projects that has been provided in creation of structure data types.

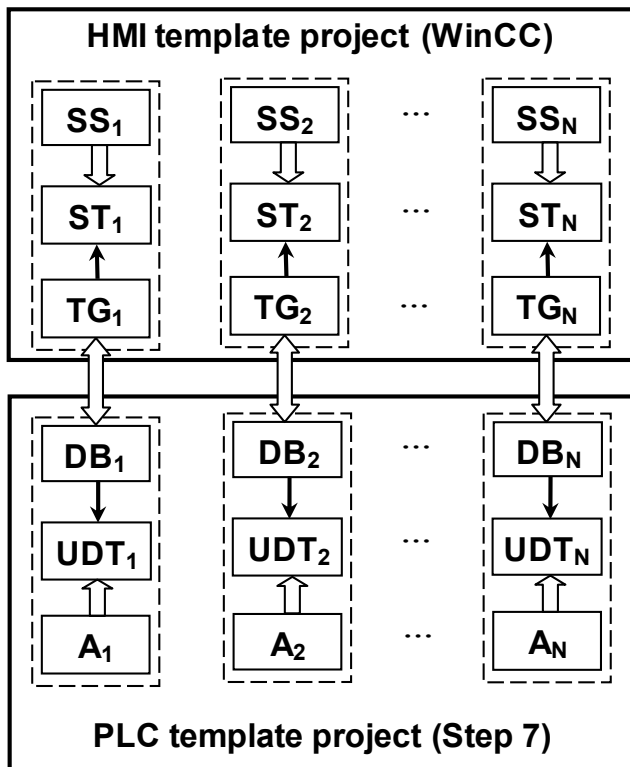


Fig. 1. Pattern structure of automated control system software:  $SS_k$  – Status Screen;  $ST_k$  – Structure Type;  $TG_k$  – Tag Group;  $DB_k$  – Data Block;  $UDT_k$  – User-Defined Type;  $A_k$  – Algorithm;  $N$  – number of types of sensors or actuators,  $k=(1...N)$ .

The second condition for a correct network data exchange is a corresponding of structure tag of HMI project to the element of the array in data block of controller user program that has been provided at the stage of data storage. The data exchange in industrial control system network is schematically shown in Fig. 1.

### D. Functionality

Functionality of same type objects of template projects includes automatic control of industrial equipment, alarm at critical state of process, monitoring and operator control of equipment.

In the PLC template project, the data processing of same type sensors and actuators, as well as the calculation of control actions performed by using the corresponding arrays created in the project. For each group of same objects, the  $A_k$  data processing algorithm is implemented in the form of a function or function block.

The controller user program is implemented by means of programming language SCL (Structured Control Language), as it allows you to implement complex algorithms and handling large quantities of structured data. Compared with other languages (LAD, FBD, STL, SFC), supported by the Step 7, on this language, it is simple enough to organize the processing of elements in the array by means of cycle operators. The functions and function blocks are called in an organizational block of the cyclic interrupt. The time interval is set appropriately to response time of occurring industrial processes.

In the HMI template project the functionality of same type objects for monitoring and operator control of industrial process, for adjustment of the settings of the equipment at the time of commissioning, as well as for tag logging and alarm logging is implemented. The graphic part of the HMI template project is a hierarchical system consisting of a base process screen (objects screen) and subordinate process screen (status screen).

Objects screen contains graphic objects of typical sensors and actuators. The current state of the objects displayed in a variety of visual effects. This is achieved through the dynamization of objects by means of dynamic dialog, or by cyclic processing of VB-scripts and C-scripts.

For each type of sensor or actuator the subordinate process screens are created. These status screens are able to display the values of input and output signals of the object in real time, the trends of signals and alarms. The status screen also allows to adjust the settings of the selected object in commissioning operations, on-line diagnosis of the process and operator control.

### E. The procedure of using a template project

The development of software based on template projects for a specific automation object begins with an analysis of the manufacturing scheme and equipment specifications. Depending on the number of sensors and actuators in the PLC project the number of elements in array in the data block  $DB_k$  for each type is determined. In the data block initialization section the initial values are assigned for certain elements of the structure type.

HMI project configuration for a specific automation object starts with a copy of the template project by means of the utility Project Duplicator. In this case, process screens and scripts of the template project are duplicated to developing

software for operator control and monitoring. Next, structure tags  $T_{km}$  in the required quantity are being created. For each structure tag the data block number and offset within that block are being determined.

The base screen of process for the particular automation object is developed as a mimic diagram of the industrial process. This is carried out by copying the graphic objects of typical sensors and actuators from the objects screen. The properties of graphic objects and their associated scripts go to the base screen of the process unchanged. On the mimic diagram of process all objects of control system and their main characteristics are displayed. Industrial control system software development based on the template project is being completed by means of the configuration of objects of process screen.

It should be noted that separation of project objects by type makes easily implement automated objects configuration of HMI project. Automated configuration can be implemented as a VBA application. In this application a group of single type objects is represented in the form of an array. Cyclic processing of array elements allows creating and configuring screen objects, archive tags and alarm messages. If a list of in use sensors and applicators change during starting-up and adjustment, this system will perform automated configuration of project objects regardless of their quantity.

#### IV. PRACTICAL USE

The proposed method of template design was implemented for the development of the software of the automated control system of oil production. For automation of oil production a large number of sensors and actuators, such as analog sensors, valves, pumps, fluid and gas meters is used.

At the same time, many oil production facilities (preliminary water discharge unit, booster pump station, gas-turbine for combustion, etc. [14 – 16]), the sensor data processing algorithms and calculation of control actions for the actuators are the same. From the viewpoint of the automation, these objects differ from each other in the number of sensors and actuators. This makes it advisable to use template projects for the development of automated control system for industrial processes software.

As an example, let's consider the creation of objects of template projects for analog sensors due to their wide use in the automation of oil production. The data obtained from different temperature, pressure and fluid level analog sensors are handled in the controller user program based on the unified algorithm.

To describe the characteristics of analog sensors in Step 7, the user-defined type  $UDT_1$ , the elements of which describes the technical settings of the sensor, the main of which are represented in Table I is created.

| Sensor settings          | Name of element | Address of element | Type of element in Step 7 | Type of element in WinCC |
|--------------------------|-----------------|--------------------|---------------------------|--------------------------|
| Current value            | Val             | +0.0               | REAL                      | FLOAT                    |
| Minimum allowed value    | rAlarmMin       | +4.0               | REAL                      | FLOAT                    |
| Maximum allowed value    | rAlarmMax       | +8.0               | REAL                      | FLOAT                    |
| Alarm state at a minimum | bAlarmMin       | +12.0              | BOOL                      | BIT                      |
| Alarm state at a maximum | bAlarmMax       | +12.1              | BOOL                      | BIT                      |

Elements of structure type  $ST_1$  in WinCC have the same name and location as the elements of the user-defined type in Step 7. To meet the conditions of exchange of data blocks the appropriate types of elements of structure data (FLOAT and BIT in WinCC; REAL and BOOL in Step 7) are selected.

To store the settings of analog sensors in the Step 7 data block  $DB_1$ , which is a one-dimensional array  $An [1..M]$  of user-defined objects such as  $UDT_1$  is created. The number of elements in the array is equal to the number of  $M$  analog sensors in project. The human machine interface software, the same amount of structure tags  $T_{1m}$ , where  $m = (1 \dots M)$  is generated. All these structure tags combined into  $TG_1$  tag group. The address of each structure tag is determined by the option of the data block  $DB_1$  and the offset of the corresponding element of the array within that block. As a result, the structure  $T_{1m}$  tag corresponds to an element of the array  $An [m]$  in the data block that provides the correct network exchange.

Analog signal processing algorithm  $A_1$  in PLC software is implemented as a function. After primary of analog sensor data, the value Val is displayed in the base screen of the process. In the processing functions there are also implemented the test of belonging of values "Val" to valid values of "rAlarmMin" and "rAlarmMax". These values are entered in the status screen  $SS_1$  in the process of commissioning. In going beyond the permissible values one of the bits, "bAlarmMin" or "bAlarmMax", is set which is reflected in the base screen by the visual effects and the alarm.

#### V. CONCLUSION

The developed method greatly simplifies the process of software development of automated control system for industrial processes with a large number of same elements of automation.

In the template projects created on the basis of this method, the structure and functionality of the industrial control system software are predetermined. The proposed structure of the template project makes it easy to adapt them to the requirements of a particular control system.

The developed method of pattern-design has proven its effectiveness in the development of automated control system for industrial processes software for a variety of oil production

TABLE I. THE MAIN ELEMENTS OF STRUCTURE TYPE FOR ANALOG SENSOR

facilities. Universality of approaches makes this method valid in the automation of various industrial facilities.

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